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Course: Ethical Hacking

Course code: ITA1407.

1) Introduction

Nmap is a network scanning tool used to identify live hosts, open ports, running services, O.S, service version. The scan report for the target 192.168.1.10 shows port 22, 80 & 3306 are open.

Analysis of open port

- *> It provides web services to users over the HTTP protocol.
- *> Data transmitting without encryption.
- *> website defacement, session hijacking.

SSH

- *> Allows secure remote administration of system.
- *> weak password may lead to brute force attack, outdated SSH software may contain vulnerabilities.
- *> Password enable SSH Key-based authentication.

MySQL

- *> provides database services for application.
- *> weak database passwords
- *> Database theft
- *> Allow database access only from trusted hosts.

Recommended Security Measure

*> Enable a firewall to restrict ports

*> Use HTTPS instead of HTTP

*> Use strong password

*> Disable unused services.

*> Monitor system logs regularly

2) Analysis of Employee

1) 40% of employees clicked the phishing link

*> Employees failed to identify and trusted unknown links.

2) 20% of Entered their Login credentials.

*> Employees entered usernames to company systems, data theft and privilege.

3) user Downloaded malicious

security weakness Identified

*> Lack of phishing awareness

*> Poor email verification practices.

*> Weak password security habits

Recommended email security controls

*> Enable spam and phishing email filters

*> Block malicious attachments & suspicious links.

*> Use antivirus software to scan email attachments.

- *? Reduced risk of phishing attacks
 - *? Better employee awareness of cyber threats.
 - *? Improve protection of information.
 - *? Stronger overall organizational security.
 - *? Faster detection.
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3? Analysis of Report

- *? Internal hostnames
- *? Mail server records
- *? DNS records
- *? Internal network structure
- *? public facing systems.

Security Implications

- *? Exposure of Internal Hostnames
- *? Exposure of mail servers
- *? Exposure of DNS records
- *? Information disclosure.

Recommended security Measures

- *? Disable DNS zone transfers for unauthorized hosts
- *? Restrict DNS access using firewall rules.
- *? Keep DNS server software updated.
- *? Monitor DNS Logs regularly.
- *? Remove unnecessary DNS records.

Benefits

- * > prevents unauthorized information disclosure
 - * > protects internet network details.
 - * > Reduces the risk of reconnaissance attacks
 - * > Improves DNS security.
 - * > strength the overall cybersecurity posture of the organization.
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